MILITARY SPECIFICATION

BALLISTICS COMPUTER, SUBASSEMBLY: 8635370
(COMPUTER UNIT)

This amendment forms a part of Military Specification MIL-B-60123(MU) dated 4 June 1964.

Paragraph 2.1, Specifications: Delete "MIL-I-45208, Inspection System Requirements".

* Paragraph 3.4.1: Delete and substitute:

"3.4.1 Operating temperature.- The computer unit shall meet the requirements of 3.6.1 to 3.6.4 inclusive while exposed and thermally stabilized at ambient temperatures of -40° and +150°F."

Paragraph 3.4.2: Delete "3.5 to 3.6.6" and substitute "3.5 to 3.6.8".

Paragraph 3.4.3: Delete and substitute:

"3.4.3 Sealing.- When pressurized to an internal pressure of 5 ± 0.1 psig, the computer unit shall not exhibit a pressure drop greater than 0.2 psig for a period of at least 2 minutes."

Add new paragraph 3.4.4 as follows:

"3.4.4 Storage temperature.- The unit shall show no evidence of physical failure and shall meet all requirements of this specification at standard ambient temperature of 60° to 90°F after having been exposed to and thermally stabilized at ambient temperature of -80°F and +160°F."

Paragraph 3.5.2: Delete "plus twice the rated voltage".

Paragraph 3.6.1: Delete "3.6.2 thru 3.6.4" and substitute "3.6.2 and 3.6.3".

Paragraph 3.6.4, third sentence: After "R618" add "and R619".

Add new paragraphs 3.6.7 and 3.6.8 as follows:

"3.6.7 Overvoltage protection.- The overvoltage protection circuit shall function to interrupt power to a load resistor connected across J604-D and J604-R when the D.C. input power to the computer unit exceeds 35.0 volts dc.

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3.6.8 Reverse polarity protection.- The application of 60 volts dc reverse polarity to the computer unit shall not cause more than 30 milliamperes of current to flow in a load resistor connected across J604D and J604R."

Paragraph 4.4.1: Delete and substitute:

"4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109."

* Add new paragraphs 4.1.2, 4.1.2.1 and 4.1.2.2 as follows:

"4.1.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) computer units plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.1.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.1.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection."

* Paragraph 4.2.2, Table I: Add the following characteristics:

"108.	Sealing	3.4.3	4.5.1.3
109.	Overvoltage protection	3.6.7	4.5.5.7
110.	Reverse polarity protection	3.6.8	4.5.5.8
111.	Dielectric strength	3.5.2	4.5.3
112.	Insulation resistance	3.5.3	4.5.4

* Paragraph 4.3: Delete "control tests" and substitute "special sampling".

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* Paragraph 4.3.1, last sentence: Delete "Control"; In tabulation heading delete "Control test" and substitute "Characteristic".

* Paragraph 4.3.2: Delete second sentence and substitute "Each sample shall meet the following requirements and tests, and shall then meet the requirements and tests in Table 1." Delete tabulation and substitute:

"CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304. Storage temperature (-80°F and +160°F)	3.4.4	4.5.1.4
305. Ammunition selection	3.4.1, 3.6.2	4.5.1.1, 4.5.5.2
306. Effective full charge count	3.4.1, 3.6.3	4.5.1.1, 4.5.5.3
307. Scale factor	3.4.1, 3.6.4	4.5.1.1, 4.5.5.4
308. Elevation amplifier output	3.4.1, 3.6.5	4.5.1.1, 4.5.5.5
309. Deflection amplifier output	3.4.1, 3.6.6	4.5.1.1, 4.5.5.6

NOTE: The tests for Characteristics 305 through 309 inclusive shall be conducted at -40°F and +150°F."

* Paragraph 4.3.3: Delete and substitute:

"4.3.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirement, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted items have been approved (see 4.2.3)."

Paragraph 4.4, first sentence: Delete "Measuring and Test Equipment of Specification MIL-I-45208 and all."

Paragraph 4.4.2: Item 2 description, delete "1 megohm ± 10%" and substitute "1 megohm minimum"; Item Number 3 description, delete and substitute "115 volts rms, 400 cps ± 1%, 2% maximum harmonic distortion"; Item Number 4 description, delete "1.0 volt peak to peak" and substitute "0.10 volt peak to peak".

Paragraph 4.5.1.1: Delete and substitute:

"4.5.1.1 Operating temperatures.- This test shall be conducted in conformance with the temperatures specified in 3.4.1. The test equipment shall be in accordance with the "Test Facilities" requirements specified in MIL-F-13926. While thermally stabilized at the specified temperatures, the computer unit shall be subjected to the tests specified in 4.5.5 to 4.5.5.4 inclusive to determine conformance with the requirements referenced herein."

* Paragraph 4.5.1.2: Delete last sentence and substitute: "The computer unit shall then be subjected to the test specified in Table I to determine conformance with the applicable requirements referenced therein."

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Paragraph 4.5.1.3: Delete and substitute:

"4.5.1.3 Sealing.- The pressure test equipment utilized for this test shall be capable of internally pressurizing the computer unit to the psig value specified. The pressure gauge of the test equipment shall permit measuring any variation in the internal pressure in increments of 0.1 psig. Remove the Plug MS-49005-1 from the front of the computer unit and replace it with an air valve containing a spring loaded valve stem. The pressure shall be applied to the computer unit through this valve. Pressurize the computer unit with clean dry air to the psig value specified in 3.4.3. The applied pressure shall be retained for a minimum of 2 minutes. Any evidence of leakage in the computer unit at the end of the 2 minute time period shall not exceed the tolerance specified in 3.4.3, as measured on the pressure gauge of the testing equipment. Upon completion of this test, remove the air valve and replace the Plug MS-49005-1 in the computer unit."

Add new paragraph 4.5.1.4 as follows:

"4.5.1.4 Storage temperature.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.4. The computer unit shall be exposed to each temperature extreme specified in 3.4.4 for a minimum of 5 hours. After each temperature exposure the computer unit shall be stabilized in an ambient temperature between 60° and 90°F and subjected to a visual, tactile and audible examination for evidence of physical failure. The computer unit shall then be subjected to the tests in 4.5.1.1 through 4.5.5.8 inclusive to determine conformance to 3.4.4. Caution shall be exercised during the temperature cycling to avoid subjecting the computer unit to thermal shock."

Paragraph 4.5.5.1: Delete "4.5.5.2 thru 4.5.5.4" and substitute "4.5.5.2 through 4.5.5.3".

Paragraph 4.5.5.3: Delete last sentence and substitute "For each ammo selected the E.F.C. counter shall perform as specified plus or minus one count, except that the tolerance for APDS-A1 ammunition shall be plus or minus three counts."

Paragraph 4.5.5.4: Delete and substitute:

"4.5.5.4 Scale factor.- To determine conformance to 3.6.4, connect the computer unit to the test circuits specified on Dwg. F8635370. Check each adjustment as specified on the computer unit drawing. Resistors R614 and R617 shall be adjusted within the tolerance specified in 3.6.4."

Add new paragraphs 4.5.5.7 and 4.5.5.8 as follows:

"4.5.5.7 Overvoltage protection.- The overvoltage protection circuit shall interrupt power to a 10 ohm ± 5%, 100 watt resistor connected across J604-D and J604-R when the D.C. input to pins P and R (pin P to be positive) exceeds 35.0 volts.

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4.5.5.8 Reverse polarity protection.- Application of 60 volts dc to the computer unit across pins P and R of J604, pin R to be positive, shall not cause current of more than 30.0 milliamperes to flow in a 100 ohm ± 5%, 100 watt load resistor connected across pins D and R of J604."

*The outside margins have been marked to indicate where additions to the previous amendment have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in this notation. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notation and relationship to the last previous amendment.

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